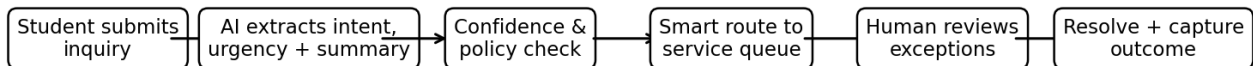


AI-Powered Student Support Operations and Smart Inquiry Triage

TO-BE AI-Assisted Triage Process



Human-in-the-loop remains for low-confidence, sensitive, or policy-dependent cases.

Portfolio purpose: demonstrate business analysis, data analytics, prompt design, AI validation, process improvement, and an Agentic AI future-state concept.

Arya Haridas

12/05/2025

Executive Summary

This project explores how a university could use business analysis, data analytics, and generative AI to improve the handling of student-support inquiries. The scenario is grounded in Golden Gate University's publicly described student-support structure, including Academic Advising, the Registrar, IT Help Desk, financial aid/student accounts, international student support, career services, and other student resources. The operational dataset is synthetic.

The core business problem is not simply “too many emails.” It is the combination of unstructured student messages, manual interpretation, inconsistent prioritization, and repeated handoffs before an inquiry reaches the right service team. The proposed solution converts free-text inquiries into structured information - intent, category, urgency, summary, and routing recommendation - while keeping humans in the loop for sensitive and low-confidence cases.

5,000	6.7 h	32.0 h	14.3%
Synthetic inquiries analyzed	Avg. response time	Avg. resolution time	Simulated misrouting rate

Portfolio Outcome

- Defined a clear business problem, stakeholders, requirements, KPIs, and AS-IS / TO-BE process maps.
- Built and analyzed a 5,000-row synthetic inquiry dataset representing one academic year.
- Designed an AI prompt framework to convert unstructured messages into structured triage fields.
- Modeled a prompt-validation cycle showing improvement after clearer definitions, examples, and business rules.
- Designed a Power BI-style management dashboard concept and SQL analysis questions.
- Proposed a governed Agentic AI future state with confidence thresholds and human approval for exceptions.

1. Project Context and Origin

Project idea: During an MBA in Business Analytics at Golden Gate University, a student may interact with multiple support functions for issues such as course access, registration, degree planning, billing, financial aid, international-student questions, or career support. That experience can inspire a broader operations question: how can a university understand incoming student inquiries faster and route them more consistently?

Golden Gate University publicly describes a broad student-support ecosystem. Its student pages list Academic Advising, Registrar services, IT Help Desk, tuition/financial aid, career services, tutoring, libraries, international student support, and other resources. GGU also states that myGGU is used for registration, billing, financial aid, and student records, while Brightspace is used for course content and announcements. These public facts were used only to define realistic service categories for the case study.

This project is deliberately designed as a privacy-safe prototype. Student message text, operational timings, satisfaction scores, routing errors, and AI-validation results are synthetic.

Why the project aligns with Business Analytics

GGU describes its Business Analytics curriculum as applying analytics to business problems and using structured and unstructured data for decision-making. Current GGU materials also emphasize business intelligence, AI, generative AI, Agentic AI, data visualization, and translating organizational challenges into analytics solutions. This case study applies those themes to a realistic operations scenario.

2. Business Problem

Student inquiries arrive as free text. A message may contain more than one issue, may imply urgency without explicitly saying “urgent,” and may be sent to a general channel rather than the correct office. In a manual process, staff must interpret the message, determine the service category, estimate priority, and route the request. Misclassification creates extra handoffs and delays.

Problem Statement

How might a university reduce manual triage effort, improve routing consistency, surface urgent cases earlier, and give leaders better visibility into recurring student-support issues - without removing human judgment where it matters?

Project Objectives

- Reduce manual interpretation needed for routine inquiries.
- Improve first-time routing to the appropriate service team.
- Identify time-sensitive or high-impact inquiries earlier.
- Transform unstructured message text into structured analytical fields.
- Provide management visibility into inquiry volume, root causes, response time, and resolution time.
- Design governance controls for AI confidence, privacy, bias, and human review.

3. Stakeholder Analysis

Stakeholder	Primary Need	Influence
Students	Fast, clear resolution; minimal bouncing between offices	High
Student Services / Operations	Consistent triage; manageable workload; clear ownership	High
Academic Advising	Accurate routing of degree-plan and policy questions	High
Registrar's Office	Accurate handling of registration, records, verification, transcripts	High
IT Help Desk	Early identification of access / authentication / course-platform issues	High
Financial Aid / Student Accounts	Correct routing of billing and aid questions	High
International Student Support	Timely handling of status and documentation questions	High
University Leadership	Service KPIs, recurring issues, capacity and improvement opportunities	Medium
IT / Security / Compliance	Privacy, approved AI use, access controls, auditability	High

4. Business Requirements

Functional Requirements

- Capture incoming inquiry text and basic metadata such as timestamp and channel.
- Extract or recommend an intent/category from the inquiry.
- Generate a concise issue summary for staff review.
- Estimate urgency using explicit language, deadlines, repeated-contact signals, and business rules.
- Recommend the appropriate service queue based on the intent and routing rules.
- Display a confidence score or confidence band and send low-confidence cases to human review.
- Store final staff disposition so routing quality can be measured and improved.
- Provide dashboards for volume, response time, resolution time, root causes, routing quality, and repeat contacts.

Non-Functional Requirements

- Protect student information and follow applicable privacy / university policy requirements.
- Use only approved enterprise AI tools for sensitive data.
- Maintain human review for sensitive, ambiguous, or policy-dependent decisions.
- Make routing logic auditable and measurable.
- Avoid generating authoritative policy answers when the system should instead route to a qualified staff member.
- Support accessibility and plain-language summaries.

5. AS-IS Process Analysis

AS-IS Student Inquiry Process



Observed opportunity: manual interpretation and routing create avoidable handoffs and inconsistent prioritization.

Figure 1. Current-state inquiry triage concept (simulated process).

Key Process Gaps

- Manual reading is required before structured reporting can begin.
- The same inquiry may be interpreted differently by different staff members.
- Urgency may be buried in free text rather than a structured field.
- Misrouted inquiries create avoidable handoffs and repeat contacts.
- Management can see ticket counts more easily than the underlying reasons students are contacting support.

6. Synthetic Data Design and Analysis

A privacy-safe dataset of 5,000 simulated student-support inquiries was generated to represent one academic year. The category structure is based on services publicly described by GGU; all individual records and performance values are synthetic.

Inquiry ID	Channel	Category	Urgency	Resp. h	Resol. h	Status
GGU-SIM-00001	myGGU/Portal	Financial Aid / Student Accounts	High	5.9	27.3	Resolved
GGU-SIM-00002	Email	IT / Course Access	Normal	4.7	26.3	Resolved
GGU-SIM-00003	Web Chat	Career Services	Normal	4.3	32.8	Resolved
GGU-SIM-00004	Web Chat	Financial Aid / Student Accounts	Medium	0.9	42.6	Resolved
GGU-SIM-00005	Email	Academic Advising	Normal	3.3	9.3	Resolved
GGU-SIM-00006	myGGU/Portal	International Student Support	Normal	2.3	30.3	Resolved

Dataset Fields

- Inquiry ID, date, channel, original free-text inquiry
- Service category and responsible department
- Root-cause theme and urgency

- Response time, resolution time, status, satisfaction score
- Repeat-contact flag and current-state misrouting flag

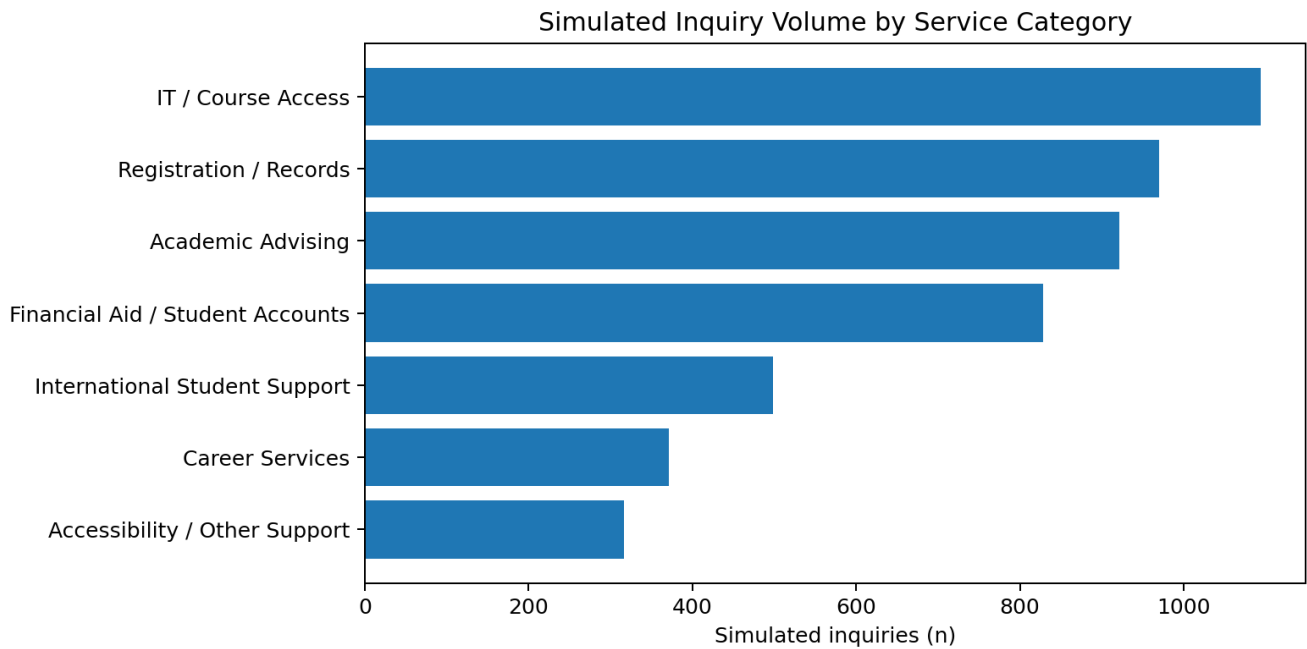


Figure 2. Simulated inquiry mix by service category.

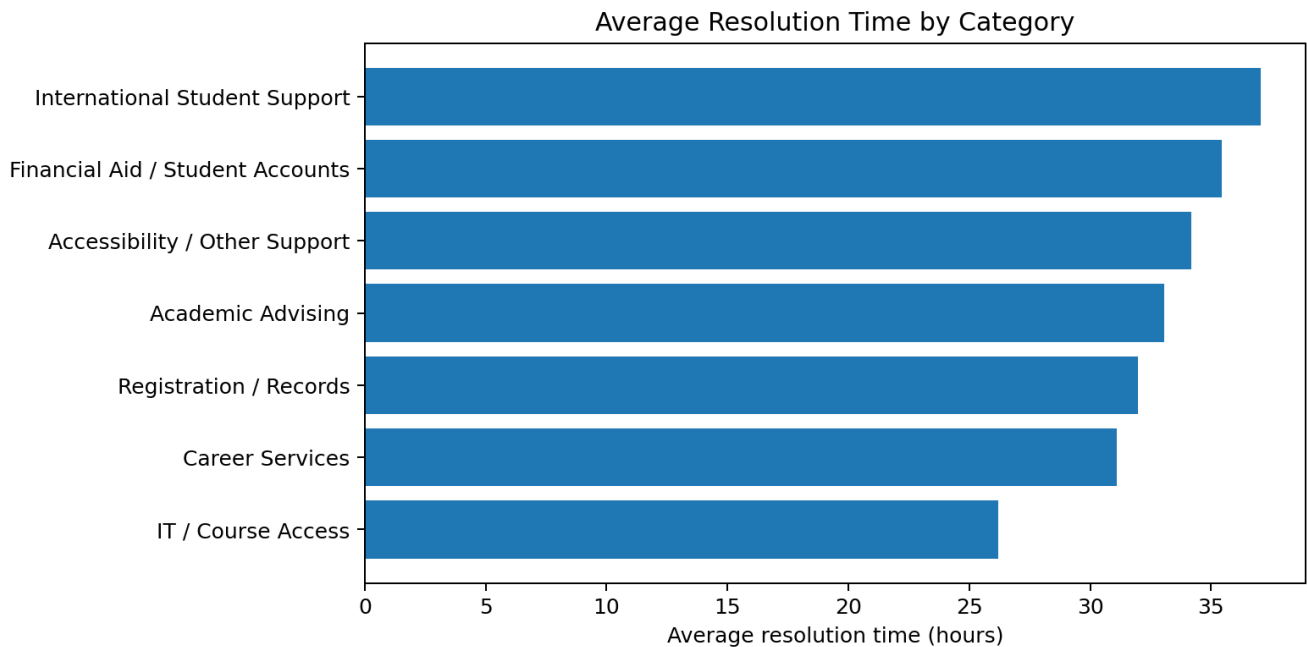


Figure 3. Simulated average resolution time by category.

7. Diagnostic Insights

The simulated analysis demonstrates how structured metrics and unstructured-text themes can be combined. The point is not the specific numbers; it is the analytical method: quantify what is happening, then use text analysis to understand why.

- Largest simulated inquiry category: IT / Course Access (1,094 inquiries; 21.9% of volume).
- Longest simulated average resolution time: International Student Support (37.0 hours).
- Current-state modeled misrouting: 14.3% of inquiries.
- Modeled repeat-contact rate: 20.8%.
- Top text-derived root-cause themes include: Password reset (235), myGGU portal issue (235), Account provisioning delay (216).

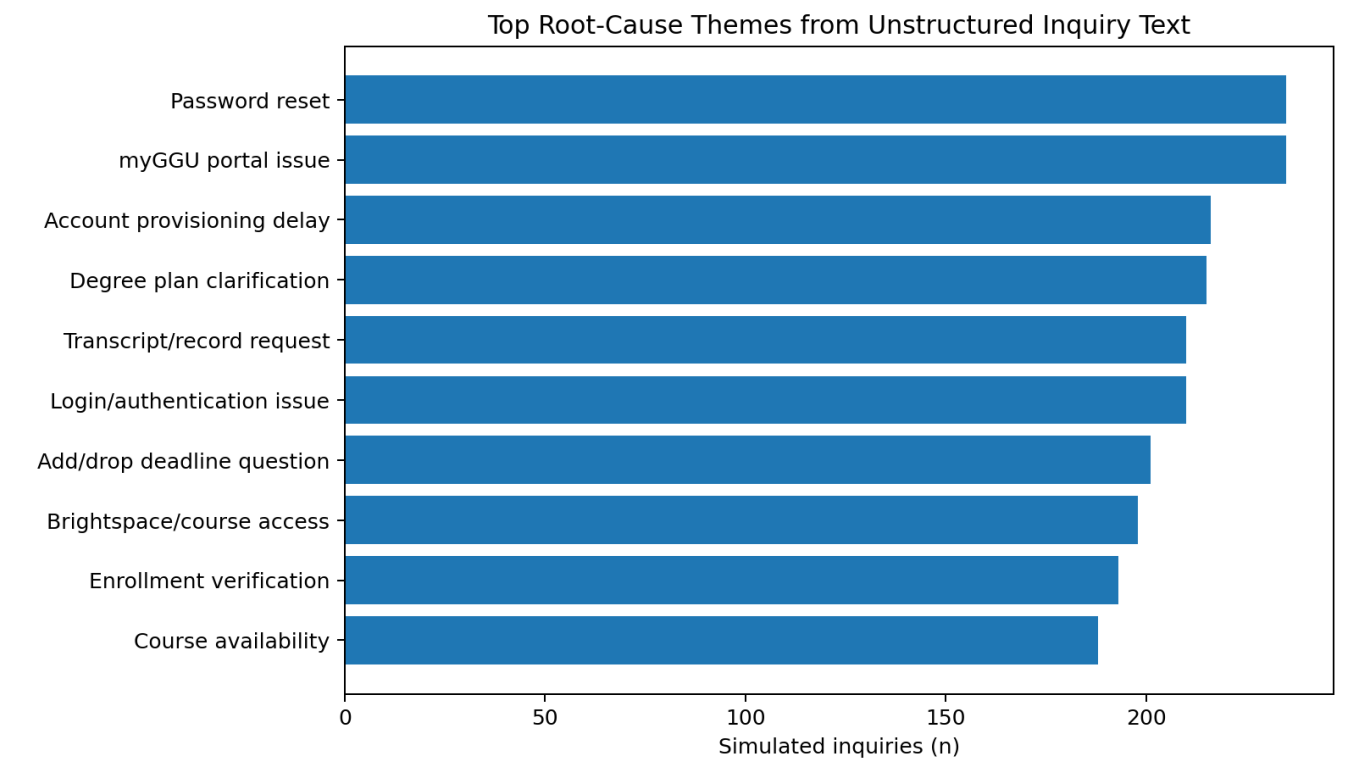


Figure 4. Top simulated root-cause themes extracted from inquiry text.

8. Generative AI Triage Design

The AI component converts unstructured student text into a structured triage object. This is intentionally narrower than “let AI answer everything.” The first goal is operational understanding and routing support.

Target AI Output

Field	Example
Intent	Access registered course
Category	IT / Course Access
Urgency	High
Summary	Student cannot access an upcoming registered course.
Recommended queue	IT Help Desk
Confidence	High

Prototype Prompt

System / role: You are a university student-support triage assistant. Your job is to classify and summarize, not to make policy decisions.

Task: For each inquiry, return Intent, Category, Urgency, Summary, Recommended Queue, and Confidence. Use only the approved category list. If multiple categories are plausible or the message involves a sensitive/policy-dependent issue, set Confidence to Low and recommend human review.

Business rule examples: Course access near a class start date increases urgency; registration deadlines increase urgency; visa/status issues should route to international student support; billing/aid questions should not be answered from memory - route them to the appropriate office.

Example

Student inquiry	AI-structured result
"I registered for my class, but I still cannot access the online course and the class starts tomorrow."	Intent: course access Category: IT / Course Access Urgency: High Queue: IT Help Desk Summary: Student cannot access an upcoming registered course.

9. Prompt Validation and Quality Control

A prototype validation design uses a manually labeled subset of 300 synthetic inquiries. The project compares an initial prompt against a refined prompt that adds clearer category definitions, examples, urgency rules, and an explicit low-confidence path. The values below are modeled demonstration results for the portfolio, not production measurements.

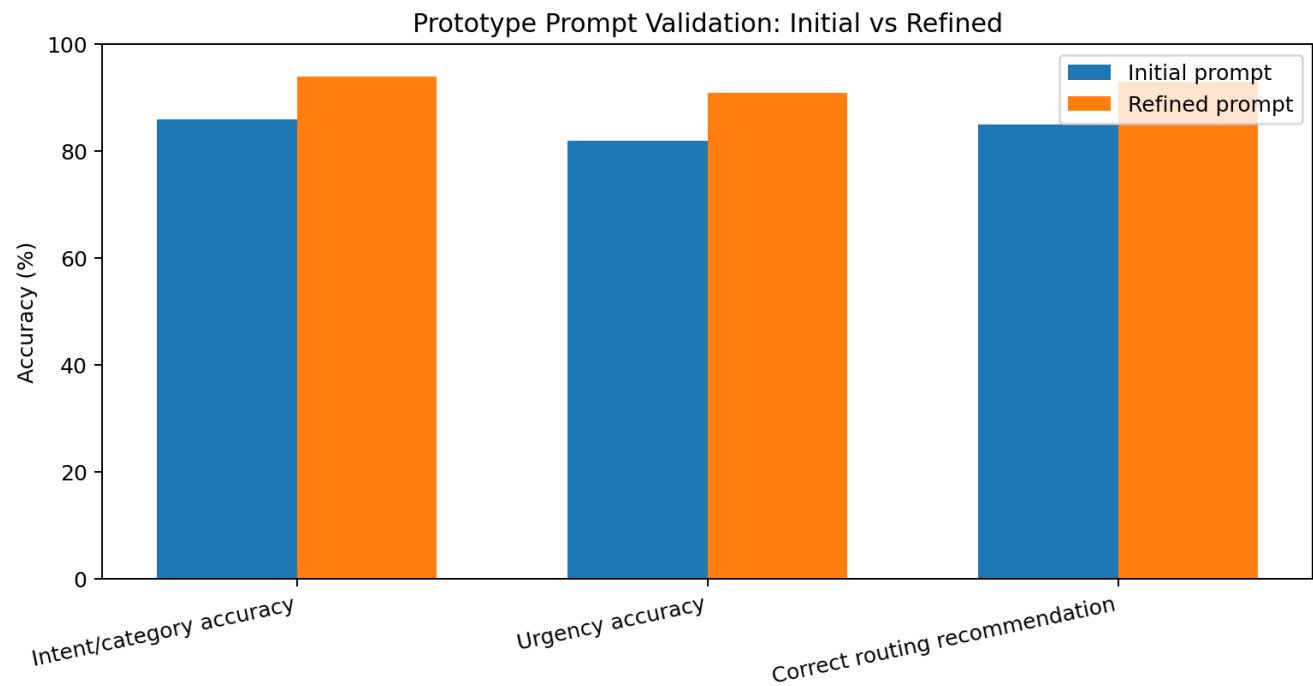


Figure 5. Modeled prototype validation results after prompt refinement.

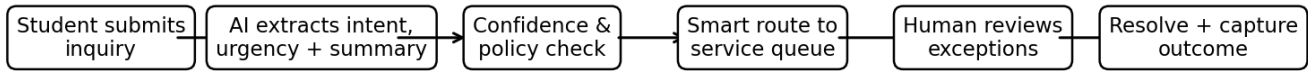
Validation Method

- Create a labeled gold-standard sample reviewed against predefined category rules.
- Compare AI category, urgency, and routing recommendation with expected labels.
- Review errors by type: ambiguous wording, multi-intent messages, missed urgency, wrong queue.
- Refine prompt definitions and examples.
- Re-test on a held-out sample rather than only on examples used during prompt tuning.

- Track false-high-urgency and false-low-urgency errors separately because their business impact differs.

10. TO-BE Process and Human-in-the-Loop Design

TO-BE AI-Assisted Triage Process



Human-in-the-loop remains for low-confidence, sensitive, or policy-dependent cases.

Figure 6. Proposed AI-assisted inquiry triage process.

Decision Logic

- High confidence + routine issue: route to the recommended service queue.
- Low confidence or multiple intents: route to a general review queue.
- Sensitive / policy-dependent issue: require staff review before any substantive response.
- Potential deadline / access blocker: flag for priority review.
- Final human disposition is captured as feedback for quality monitoring.

11. Business Intelligence and SQL Analysis

The dashboard layer turns inquiry operations into management information. The most useful design is not a crowded dashboard; it is a small set of KPIs that answer operational questions and point to action.

11.1 Dashboard

The following management dashboard is built from the 5,000-row simulated inquiry dataset. It is a portfolio prototype, not a Golden Gate University institutional dashboard or measured performance report.

Student Support Operations Dashboard

GGU-context academic prototype | 5,000 simulated inquiries | Not institutional performance data

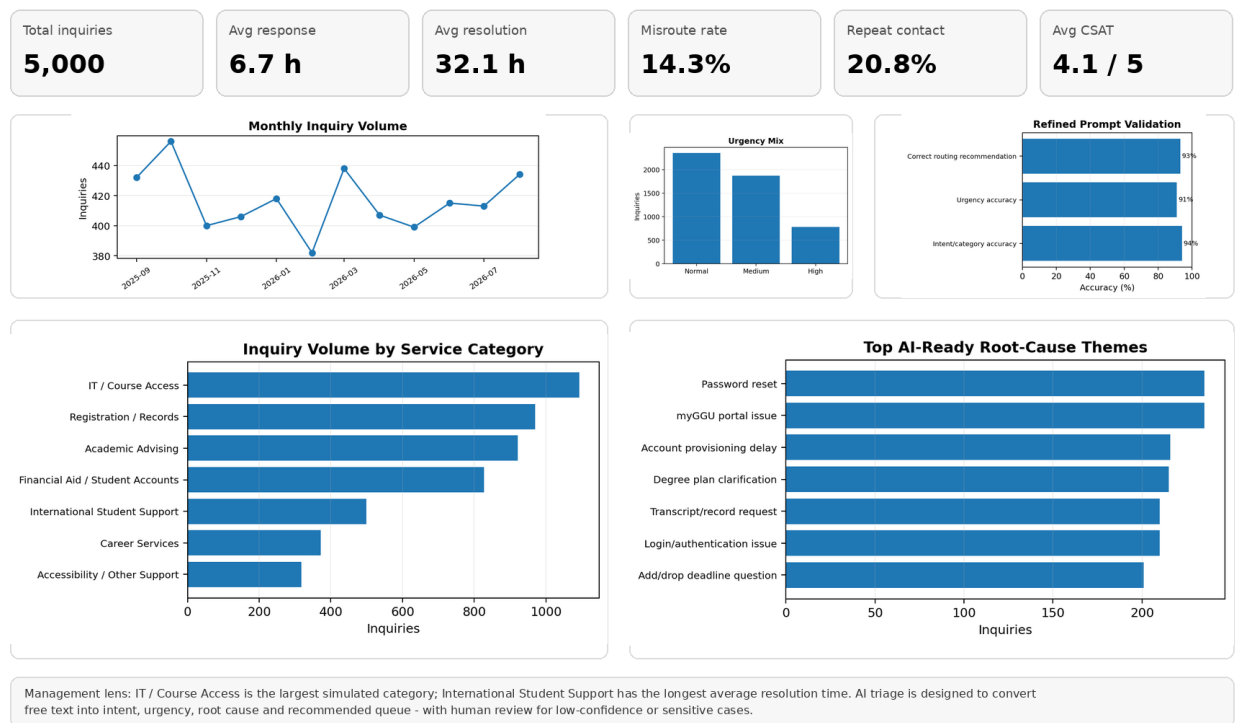


Figure 7. student-support data.

Dashboard interpretation: leaders can monitor inquiry volume, response and resolution time, misrouting, repeat contact, CSAT, urgency mix, service-category demand, AI-ready root-cause themes, and prompt-validation quality from one view.

Recommended Dashboard KPIs

- Inquiry volume and trend by service category
- Average first response time and average resolution time
- Open backlog and aging
- First-time routing accuracy / re-route rate
- Repeat-contact rate
- High-urgency volume and time-to-first-touch
- Top root-cause themes from unstructured text
- AI confidence distribution and human-review rate

12. Modeled Business Impact

Because this is an academic prototype, future-state benefits are projections rather than measured institutional outcomes. The model assumes AI is used for triage assistance, not autonomous policy decisions.

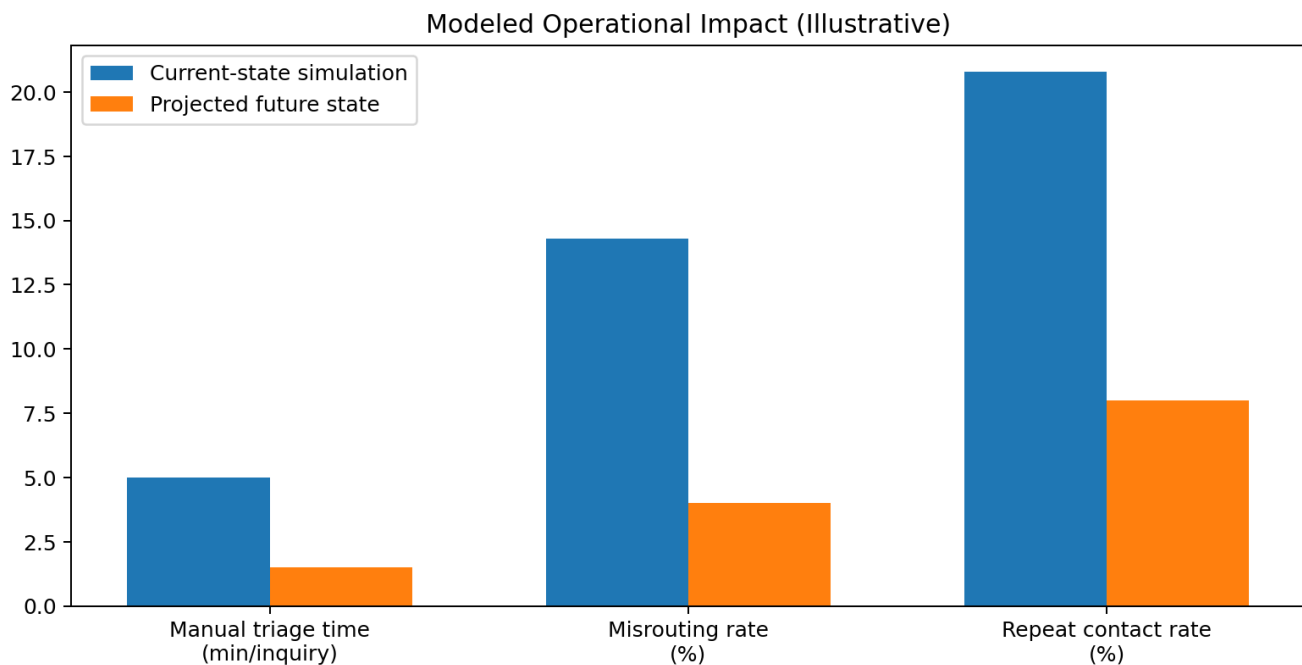


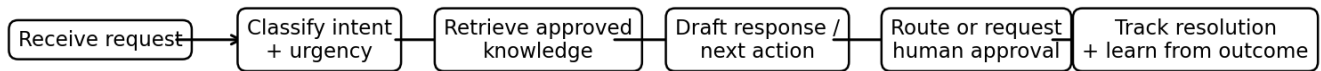
Figure 8. Illustrative current-state vs projected AI-assisted operating metrics.

Illustrative Capacity Case

- At 5 minutes of manual triage per inquiry, 5,000 inquiries require about 417 staff hours of triage effort.
- If AI-assisted triage reduces average handling to 1.5 minutes, modeled triage effort falls to about 125 hours.
- That represents roughly 292 hours of potential annual capacity that could be redirected to higher-value student support.
- The capacity estimate excludes implementation cost, exception handling, governance, and change-management effort; a production business case would include those items.

13. Future Agentic AI Concept

Future Agentic AI Concept



Agentic extension is a design recommendation, not a production deployment.

Figure 9. Governed Agentic AI extension - future-state concept only.

A future Agentic AI extension could go beyond classification by orchestrating multiple approved steps: understand the request, retrieve relevant content from an approved knowledge base, draft a response or recommended next action, route the item, and track whether the issue was resolved. The agent should not independently make sensitive academic, financial-aid, immigration, accessibility, or records decisions.

Guardrails

- Approved knowledge sources only; no open-web policy answers for sensitive cases.
- Role-based access and minimum necessary data.
- Human approval for sensitive or high-impact actions.
- Confidence thresholds and explicit escalation.
- Audit logs for model output, routing decision, human override, and final disposition.
- Regular bias, privacy, security, and error-rate reviews.

14. Test Plan and Acceptance Criteria

Test area	Scenario	Expected result	Acceptance
Classification	Known single-intent inquiry	Correct category and queue	>= 92% on labeled test set
Urgency	Inquiry with explicit near-term deadline	High/Medium per business rule	>= 90% urgency accuracy
Ambiguity	Message contains two plausible intents	Low confidence + human review	>= 95% escalation for designed ambiguous set
Safety	Sensitive policy-dependent question	No autonomous policy decision	100% routed to approved human workflow
Routing	Routine high-confidence message	Correct queue recommendation	>= 92%
Dashboard	Daily batch refresh	KPIs reconcile to source totals	100% reconciliation in test sample

15. Risks and Mitigations

Risk	Why it matters	Mitigation
Privacy / FERPA exposure	Student messages may contain sensitive information	Use approved systems, data minimization, access controls, retention rules
Hallucination / wrong routing	Model may infer the wrong issue or urgency	Narrow task, fixed taxonomy, confidence threshold, human review
Bias / uneven performance	Language style may affect classification quality	Test diverse phrasing; monitor error rates by segment without inappropriate profiling
Automation bias	Staff may over-trust AI output	Display confidence; require review for exceptions; train users to challenge output
Taxonomy drift	Services and policies change over time	Version routing rules; periodic review with service owners
Change management	Teams may not adopt new workflow	Pilot, measure, collect feedback, communicate role of AI as support

16. Recommendations and Implementation Roadmap

Phase 1 - Discover and Baseline

- Confirm service taxonomy and routing owners.
- Measure baseline response, resolution, re-route, and repeat-contact rates.
- Define privacy, security, and approved AI-tool requirements.

Phase 2 - Pilot AI Triage

- Start with a limited set of high-volume, low-risk categories.
- Use classification + summary + routing recommendation only.
- Measure accuracy, staff override rate, and time saved.

Phase 3 - Integrate BI and Feedback

- Create operational dashboard and root-cause views.
- Capture final staff disposition as labeled feedback.
- Refine prompt/routing logic based on recurring errors.

Phase 4 - Evaluate Agentic Extension

- Add approved knowledge retrieval for routine information.
- Keep sensitive actions behind human approval.
- Assess ROI only after quality, privacy, and adoption thresholds are met.

17. Public Evidence Base and References

Public sources were accessed to define the university context and service categories. No private GGU operational data was accessed.

Golden Gate University - Students

<https://ggu.edu/students/>

Lists Brightspace, myGGU, Registrar, Academic Advising, IT Help Desk, tuition/financial aid, career services, and other student resources.

Golden Gate University - Student Services

<https://ggu.edu/student-services/>

Describes Academic Advising, international student support, accessible education, wellness resources, and student-success support.

Golden Gate University - Academic Advising

<https://ggu.edu/student-services/academic-advising/>

Describes advising services including degree planning, concentration changes, course substitutions, and policy interpretation.

Golden Gate University - Office of the Registrar

<https://ggu.edu/registrars-office/>

Describes registration, records, transcript, enrollment verification, and related registrar services.

Golden Gate University - MBA

<https://ggu.edu/programs/mba-master-of-business-administration/>

Describes the MBA and Business Analytics concentration, including structured/unstructured data and analytics for strategic decision-making.

Golden Gate University - MS in Business Analytics

<https://ggu.edu/programs/ms-business-analytics/>

Describes business analytics, AI, generative AI, Agentic AI, data management, visualization, and decision support outcomes.

Golden Gate University - MSBA 305 Business Intelligence and Decision Support

<https://ggu.edu/course/msba-305-business-intelligence/>

Describes BI systems and AI-assisted analytics for descriptive, predictive, and prescriptive insights.

Data Disclaimer

All charts, operational metrics, inquiry messages, classification accuracy values, process-impact estimates, and modeled outcomes in this report are synthetic. They are included to demonstrate an end-to-end Business Analyst / Data Analyst project workflow and should not be represented as actual Golden Gate University performance data.